

Imperial College London

MSc EXAMINATION June 2025

This paper is also taken for the relevant Examination for the Associateship

STATISTICAL METHODS FOR EXPERIMENTAL PHYSICS

For MLBD MRes Students

June 4th, 2025: 10:00 to 12:00

Answer all FOUR questions

Marks shown on this paper are indicative of those the Examiners anticipate assigning.

General Instructions

Complete the front cover of each of the 4 answer books provided.

If an electronic calculator is used, write its serial number at the top of the front cover of each answer book.

USE ONE ANSWER BOOK FOR EACH QUESTION.

Enter the number of each question attempted in the box on the front cover of its corresponding answer book.

Hand in 4 answer books even if they have not all been used.

You are reminded that Examiners attach great importance to legibility, accuracy and clarity of expression.

1. The exponential decay of a radioactive isotope is described by the exponential probability density function (PDF) given by

$$f(t) = \lambda e^{-\lambda t} \quad (t \geq 0),$$

where t is the time of decay of the isotope and $\lambda > 0$ is the decay constant.

- (i) Show that the expectation value of the decay time t is given by

$$E[t] = \frac{1}{\lambda}.$$

HINT: You may find the following integral useful:

$$\int_0^{\infty} x e^{-\alpha x} dx = \frac{1}{\alpha^2} \quad (\alpha > 0).$$

ANSWER: [SEEN] : The expectation value of the decay time t is given by

$$E[t] = \int_0^{\infty} t f(t) dt.$$

[1]. Substituting the exponential PDF into the integral, we have,

$$E[t] = \int_0^{\infty} t \lambda e^{-\lambda t} dt = \lambda \frac{1}{\lambda^2} = \frac{1}{\lambda},$$

where we've used the hint to evaluate the integral [1]. [2 marks]

- (ii) Using the method of moments, find an estimator of the decay constant λ from a set of N independent measurements of the decay time t_i , where $i = 1, 2, \dots, N$.

ANSWER: [SEEN] : The method of moments estimator just says to take the expectation value of the decay time t and set it equal to the sample mean [1],

$$E[t] = \frac{1}{\lambda} = \frac{1}{N} \sum_{i=1}^N t_i.$$

[1]. Solving for λ gives,

$$\hat{\lambda} = \frac{N}{\sum_{i=1}^N t_i}.$$

[1]. [3 marks]

- (iii) If t_1 and t_2 are the decay times of two isotopes with the same decay constant λ , show that the probability of observing $T = t_1 + t_2$ is given by

$$f(T) = \lambda^2 T e^{-\lambda T} \quad (T \geq 0)$$

if the decay times are independent.

ANSWER: [UNSEEN] : Students are expected to know how to derive simple

[This question continues on the next page ...]

pdf compositions, and have explicitly seen a calculation of the sum of two independent random variables. We start by using the standard change of variables,

$$T = t_1 + t_2 \quad V = t_1$$

[1]. The Jacobian of the transformation is simple and,

$$dt_1 dt_2 = dT dV.$$

The marginal pdf of an independent sum of random variables T is given by

$$f(T) = \int_{-\infty}^{\infty} f(V)g(T-V)dV.$$

[1]. Substituting the exponential pdfs of the two independent random variables, we have,

$$\begin{aligned} f(T) &= \int_0^T \lambda e^{-\lambda V} \lambda e^{-\lambda(T-V)} dV \\ &= \lambda^2 \int_0^T e^{-\lambda(V+T-V)} dV \\ &= \lambda^2 e^{-\lambda T} \int_0^T 1 dV \\ &= \lambda^2 T e^{-\lambda T}. \end{aligned}$$

[2]. In order to get the correct answer, we needed to use the fact that since $T \geq 0$, we must have that $V = t_2 \leq T$, hence the upper limit of the integral is T and not ∞ [1]. [5 marks]

Find the maximum likelihood estimate of the decay constant λ from a set of N independent measurements of the sum of decay times T_i , where $i = 1, 2, \dots, N$.

ANSWER: [UNSEEN] (very similar to simple exponential case): The likelihood function is just the product of the pdf over each of the data,

$$L(\lambda) = \prod_{i=1}^N f(T_i) = \prod_{i=1}^N \lambda^2 T e^{-\lambda T_i}$$

[1]. To find the maximum likelihood estimate, we take the logarithm of the likelihood function and differentiate with respect to λ and set the result to zero,

$$\ln L(\lambda) = \sum_{i=1}^N \ln f(T_i) = 2N \ln \lambda + \sum_{i=1}^N \ln T_i - \lambda \sum_{i=1}^N T_i.$$

[1]. Differentiating with respect to λ gives,

$$\frac{d \ln L}{d \lambda} = \frac{2N}{\lambda} - \sum_{i=1}^N T_i = 0.$$

[This question continues on the next page ...]

[1]. Solving for λ gives,

$$\hat{\lambda} = \frac{2N}{\sum_{i=1}^N T_i}.$$

[1]. [4 marks]

- (iv) Show that the maximum likelihood estimate of the decay constant λ that you derived in the previous question is a biased estimator. HINT: You may find the following inequality useful:

$$E\left[\frac{1}{\sum x}\right] \geq \frac{1}{E[\sum x]}.$$

ANSWER: [UNSEEN] : The definition of an unbiased estimator is one for which, $E[\hat{\lambda}] = \lambda$ **[1].** The expectation value of our maximum likelihood estimate of the decay constant λ is given by

$$E[\hat{\lambda}] = E\left[\frac{2N}{\sum_{i=1}^N T_i}\right] \geq \frac{2N}{\sum_{i=1}^N E[T_i]} = \frac{2N}{NE[T]} = \frac{2}{E[T]},$$

where we have used the hint given **[2].** Next, we need to find the expectation value of the sum of the decay times T , which is given by

$$E[T] = E[t_1 + t_2] = E[t_1] + E[t_2] = 2E[t] = \frac{2}{\lambda}.$$

[2] by the linearity of expectation **[1].** Substituting this into the previous equation gives,

$$E[\hat{\lambda}] \geq \frac{2}{2/\lambda} = \lambda.$$

[1]. Hence, we have shown that the maximum likelihood estimate of the decay constant λ is a biased estimator [1]. [8 marks]

- (v) Typically, *good* estimators are unbiased. Name two other properties of good estimators, and describe what they mean.

ANSWER: [SEEN] : Students can choose any two of the following properties of good estimators, or an alternative property, and describe them. :

- **Consistency:** A consistent estimator is one for which the estimate converges to the true value of the parameter as the sample size increases. This means that as we take more and more measurements, the estimate becomes more accurate.
- **Efficiency:** An efficient estimator is one that has the smallest possible variance among all unbiased estimators. This means that it provides the most precise estimate of the parameter.
- **Simple to evaluate:** A good estimator should be easy to compute and not require complex calculations or assumptions. This makes it more practical to use in real-world situations.

[1] mark for 2 properties, and **[2]** marks for the descriptions. [3 marks]

[Total 25 marks]

2. The Poisson probability distribution for a discrete random variable k is given by

$$p(k; \lambda) = e^{-\lambda} \frac{\lambda^k}{k!}$$

- (i) Show that the maximum likelihood estimator for the parameter λ , assuming an observed value of k is given by

$$\hat{\lambda} = k.$$

ANSWER: [SEEN] : From the definition of the Poisson distribution, we find

$$L(\lambda) = \frac{\lambda^k}{k!} e^{-\lambda}$$

and so taking a negative log, we get,

$$q = -\ln L(\lambda) = \lambda - k \ln \lambda + \ln(k!)$$

[1]. The maximum likelihood estimator for λ is found by minimizing q (equivalent to maximizing L) with respect to λ [1].

$$\left. \frac{dq}{d\lambda} \right|_{\lambda=\hat{\lambda}} = 0 \implies 1 - \frac{k}{\hat{\lambda}} = 0$$

and so,

$$\hat{\lambda} = k$$

[1].

[3 marks]

- (ii) A series of n independent sequential measurements is made of the number of light pulses from a distant galactic source in a fixed interval of time. Each measurement is an integer number k_i , for $i = 1 \dots n$, which is distributed as a random Poisson variable with parameter λ_i . Figure 1 shows a set of such measurements. It is assumed that the source pulses at a constant rate of T pulses over a single measurement interval of time. Under this assumption $\lambda_i = T$ for every i .

- (a) Write down the likelihood function for the set of n measurements as a function of T , assuming a constant rate T .

ANSWER: [SEEN] : This is just the product of probabilities for each measurement [1],

$$L(T) = \prod_{i=1}^n p(k_i; \lambda_i = T) = \prod_{i=1}^n e^{-T} \frac{T^{k_i}}{k_i!}$$

[1]. 1 mark is lost for not using T in the expression.

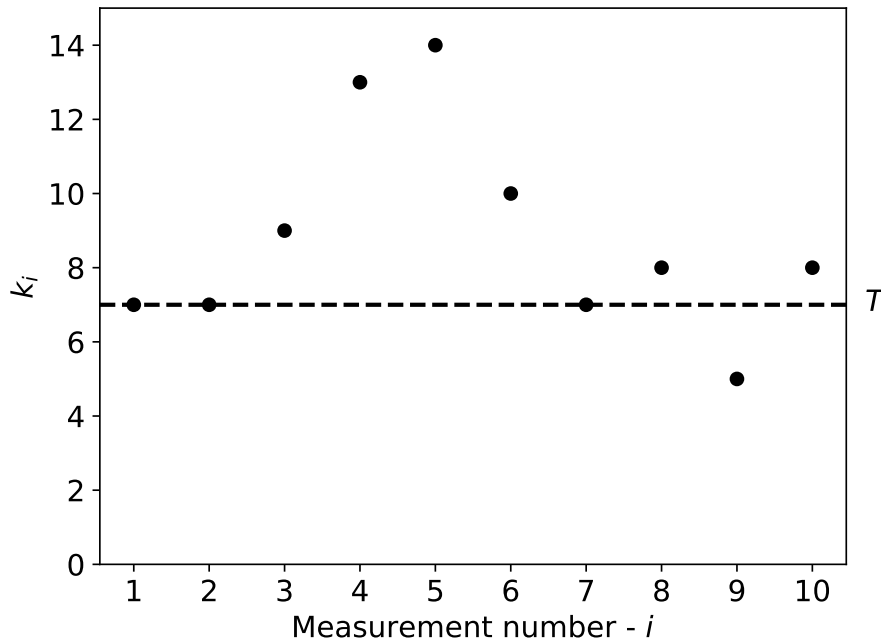


Figure 1

- (b) Find the expression for the maximum likelihood estimator $\hat{T}$ given the measurements.

ANSWER: It's simplest to first find $q = -\ln L(T)$,

$$q = -\ln L(T) = -\sum_{i=1}^n (-T + k_i \ln T) = nT - \sum_{i=1}^n k_i \ln T,$$

[1] and take the derivative with respect to T , setting to zero [1] to find $\hat{T}$,

$$0 = \frac{dq}{dT}|_{T=\hat{T}} = n - \frac{1}{\hat{T}} \sum_{i=1}^n k_i$$

[1] or re-arranging, we find,

$$\hat{T} = \frac{1}{n} \sum_{i=1}^n k_i,$$

[1] the estimator is just the average observed value of the measurements.

- (c) Describe how you would determine a 95% confidence region for the parameter T from the measurements using the Neyman construction and the profile likelihood ratio test statistic,

$$t = -2 \ln \left(\frac{L(T)}{L(\hat{T})} \right).$$

[This question continues on the next page ...]

ANSWER: [SEEN] : The answer should contain the following set of steps:

1. Calculate the value of the test statistic for the data t_{obs} , which should depend T . [1]
2. For a set of T values derive the distribution $f(t; T)$ of the test-statistic assuming T is the true value. Choose a procedure to construct 95% of the distribution in t for each value of T . Any well motivated procedure is ok, for example the likelihood ratio ordering principle is a good choice. [2]
3. For each value of T determine whether the observed value t_{obs} lies inside the 95% region or not. This could be based on calculating a p -value $p = \int_{t_{obs}}^{\infty} f(t; T) dt$ and choosing a set of points T for which $p > 0.05$. [1]
4. The union of those chosen points forms the 95% confidence interval for T . [1]

The last two points can be merged into a single step. Any test statistic can be used if justified. [11 marks]

- (iii) An alternate hypothesis for the measured pulses is that the rate of pulses of the source varies with time. Figure 2 shows the same measurements but with the two hypotheses overlaid: H_0 represents the constant rate hypothesis and H_1 represents an alternative hypothesis assuming a time varied pulse rate.

- (a) State the definition of Type-I and Type-II errors in terms of the two hypotheses H_0 and H_1 .

ANSWER: [SEEN] : Answers can be given in the form of words or equations. The Type-I error is the probability to reject the null hypothesis when the null hypothesis is true and the Type-II error is the probability to reject the alternate hypothesis (or accept the null hypothesis), when the alternate hypothesis is true.

$$\text{type-I error} = \alpha = P(X \in w | H_0)$$

[1]

$$\text{type-II error} = \beta = P(X \in W - w | H_1)$$

[1] where $X \in W$ is some test statistic and w is the critical region [2]. Answers that just reference integrals are ok provided that the region of integration is specified and not restricted to the 1D lower or upper bound integral case used in lectures for p -values. For example, it's ok to write,

$$\text{type-I error} = \int_w f(X|H_0) dX$$

$$\text{type-II error} = \int_{W-w} f(X|H_1) dX$$

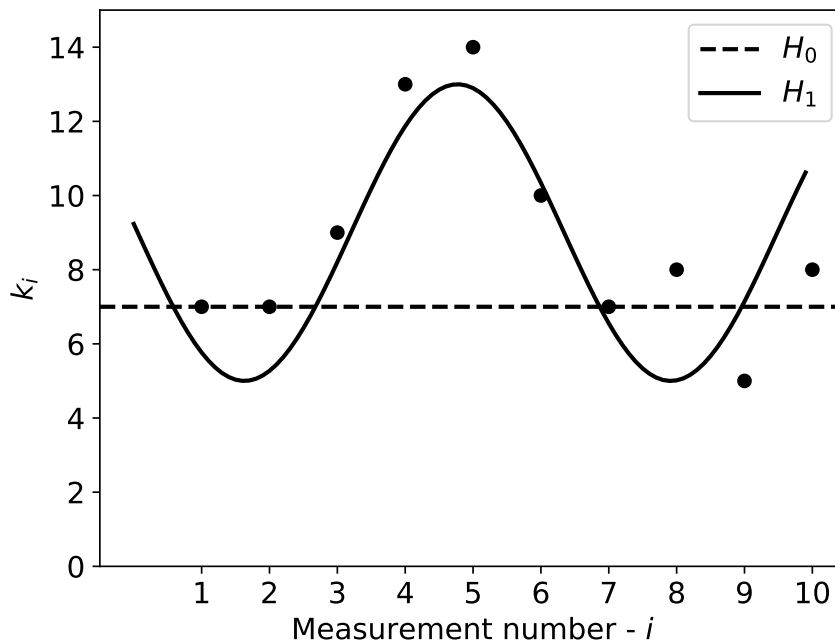


Figure 2

- (b) Devise a hypothesis test to test if H_0 is a good description of the data, based on a likelihood ratio test statistic,

$$t = -2 \ln \left(\frac{L(H_1)}{L(H_0)} \right)$$

and describe how you would perform the test using the data shown in Figure 2. You should *clearly* describe each step of the test, explaining how to generate distributions of your test statistic, and define your type-I error and how you would calculate the power of your test. You do not need to calculate or write out in full the likelihoods or the test statistic. You can use diagrams to help describe your procedures.

ANSWER: [UNSEEN] : The following steps should be included,

- 1. Given the test statistic, choose a threshold for the size of the test α , eg $\alpha = 0.05$. [1]**
- 2. The distribution of the test statistic under H_0 can be generated with random poisson toys, setting the values of $\lambda_i = T$, and one can use that to determine a critical region w such that $P(t \in w | H_0) = 0.05$. [3]**
- 3. Evaluate the observed value of the test statistic t_{obs} . If $t_{obs} \in w$ then we reject the null hypothesis - the fit is not of good quality. [1]**
- 4. The power $1 - \beta$ should be calculated as $1 - P(t \in W - w | H_1)$. Again the**

[This question continues on the next page ...]

distribution of t under H_1 can be generated with Poisson random toys.

[2]

Marks are lost for not clearly explaining how to generate distributions of test statistics, or not explaining how to go about determining the critical region.

[11 marks]

[Total 25 marks]

3. (i) (a) What is meant by a *conjugate prior*?

ANSWER: SEEN: A conjugate prior, when combined with a likelihood of given form, gives a posterior of the same mathematical form, but with different (hyper)parameters. [3]

- (b) What is the technical distinction between $p(\text{data}|\text{parameters})$ when it is treated as a sampling distribution and as a likelihood?

ANSWER: SEEN: sampling distribution: function of data with parameters fixed. Likelihood: function of parameters with data fixed. [3]

[6 marks]

- (ii) A polling organisation wishes to predict the outcome of the next election, which is being contested by two parties, A and B . Its goal is to infer the probability q that electors will vote for party A . It polls N people, who answer truthfully and independently of each other who they would vote for. n people respond that they would vote for party A .

Argue that the probability of this happening is the binomial distribution

$$p(n) = \binom{N}{n} q^n (1-q)^{N-n}.$$

ANSWER: SEEN: The probability of a single person voting for A is q and for B it is $1-q$. Independence implies multiplying the probabilities, and the prefactor accounts for the number of possible orderings that give rise to the same n . [5 marks]

- (iii) (a) In the absence of any other information, what would be a suitable prior for q ? Normalise it correctly.

ANSWER: UNSEEN: Uniform between 0 and 1, so, normalising, $\pi(q) = 1$. [2]

- (b) The polling organisation has some prior information from previous surveys, and chooses a Dirichlet prior $\pi(q) = D(q; \alpha, \beta)$, which is defined by

$$\pi(q) = D(q; \alpha, \beta) \propto q^\alpha (1-q)^\beta,$$

and α and β are (hyper)parameters of the prior. The constant of proportionality ensures that the prior is normalised, but is not relevant and can be ignored.

What values of α and β correspond to a uniform distribution?

ANSWER: UNSEEN: $\alpha = \beta = 0$. [2]

- (c) Show that the posterior is given by another Dirichlet distribution, and find expressions for the posterior parameters in terms of α, β, n and N .

ANSWER: UNSEEN: By multiplying the likelihood and the prior, by inspection $\alpha \rightarrow \alpha + n, \beta \rightarrow \beta + N - n$. [3]

[7 marks]

- (iv) (a) Show that the maximum a posteriori (MAP) value of q in this case is given by

$$q_{\text{MAP}} = \frac{\alpha + n}{\alpha + \beta + N}.$$

[This question continues on the next

ANSWER: UNSEEN:

$$\begin{aligned}
 \ln p(q) &= \text{cst.} + (\alpha + n) \ln q + (\beta + N - n) \ln(1 - q) \\
 \Rightarrow 0 &= \frac{\partial \ln p(q)}{\partial q} = \frac{(\alpha + n)}{q} - \frac{(\beta + N - n)}{1 - q} \\
 \Rightarrow 0 &= (\alpha + n)(1 - q) - (\beta + N - n)q \\
 \Rightarrow q_{\text{MAP}} &= \frac{\alpha + n}{\alpha + \beta + N}.
 \end{aligned}$$

[4]

- (b) By considering the second derivative of the log posterior, find an expression for the approximate posterior variance of q .

ANSWER: UNSEEN: Taking the second derivative, and evaluating at q_{MAP} , noting that

$$1 - q_{\text{MAP}} = \frac{\beta + N - n}{\alpha + \beta + N}.$$

$$\begin{aligned}
 \sigma_q^{-2} &\simeq -\frac{\partial^2 \ln p(q)}{\partial q^2} = \frac{(\alpha + n)}{q^2} + \frac{(\beta + N - n)}{(1 - q)^2} \\
 \Rightarrow \sigma_q^{-2}|_{q=q_{\text{MAP}}} &\simeq \frac{(\alpha + \beta + N)^2}{\alpha + n} + \frac{(\alpha + \beta + N)^2}{\beta + N - n} \\
 \Rightarrow \sigma_q^2 &\simeq \frac{(\alpha + n)(\beta + N - n)}{(\alpha + \beta + N)^3}. \tag{1}
 \end{aligned}$$

[3]

[7 marks]

[Total 25 marks]

4. (i) (a) From the laws of probability, derive Bayes theorem

$$p(X|Y) = \frac{p(Y|X)p(X)}{p(Y)}.$$

ANSWER: SEEN: By the product rule, the joint probability (density) of X and Y is

$$p(X, Y) = p(X|Y)p(Y).$$

Since $p(X, Y) = p(Y, X)$, reversing X and Y and equating gives the result. [2]

- (b) A faulty theft detection device at the exit of a department store incorrectly goes off when an innocent shopper leaves 2% of the time. It will also fail to detect a real shoplifter in 10% of cases. If 99% of shoppers are honest, use Bayes theorem to calculate the probability that a shopper is a shoplifter if the alarm goes off.

ANSWER: Variant of SEEN: Let A = alarm goes off. Let T represent ‘shopper is a thief’. We want $p(T|A)$. With Bayes theorem, the marginalisation rule (with $\sim T$ meaning not- T , i.e. innocent) and the product rule:

$$\begin{aligned} p(T|A) &= \frac{p(A|T)p(T)}{p(A)} = \frac{p(A|T)p(T)}{p(A, T) + p(A, \sim T)} \\ &= \frac{p(A|T)p(T)}{p(A|T)p(T) + p(A|\sim T)p(\sim T)} \\ &= \frac{0.9 \times 0.01}{0.9 \times 0.01 + 0.02 \times 0.99} \\ &= 0.31 \end{aligned} \tag{1}$$

[6]

[8 marks]

- (ii) (a) When comparing two models M_0 and M_1 , the posterior odds for each model, $p(M_i|d)$ (given some data d) may be calculated using Bayes theorem. Show that the relative posterior odds is equal to the Bayes factor (the ratio of the Bayesian evidence values $p(d|M_i)$) if a condition is satisfied (you should state this), and derive an expression for the Bayesian evidence involving its model parameters.

ANSWER: SEEN: We want $p(M|d)$ and Bayes theorem gives

$$p(M|d) = \frac{p(d|M)\pi(M)}{p(d)}$$

so the relative posterior odds are

$$\frac{p(M_0|d)}{p(M_1|d)} = \frac{p(d|M_0)}{p(d|M_1)} \frac{p(M_0)}{p(M_1)}$$

[3] so for equal prior model probabilities, the relative posterior odds is the ratio of the Bayesian evidence values. [2]

By marginalisation over the model parameters θ ,

$$p(d|M) = \int p(d, \theta|M) d\theta = \int p(d|\theta, M) p(\theta|M) d\theta.$$

[2]

- (b) What requirement is placed on the prior on the parameters in order to compute the Bayesian evidence?

ANSWER: SEEN: It has to be a proper probability distribution. i.e. it gives unity when integrated over the parameters. [2]

[8 marks]

- (iii) (a) A man arrives in a city and takes a random taxi with city licence number n to his hotel. Assuming that licences are issued in sequence from 1 upwards, what is the posterior probability for the total number of taxis in the city, N ? Assume a uniform prior for N (truncated at some large value), and there is no need to normalise the posterior (or the prior).

ANSWER: UNSEEN: We want $p(N|n)$. The likelihood is just $1/N$ if $N \geq n$, assuming the taxi is picked randomly. Hence

$$p(N|n) \propto p(n|N) p(N) \propto \frac{1}{N} \quad \text{for } N \geq n.$$

and zero for $N < n$. [5]

- (b) What is the maximum likelihood estimate for N ?

ANSWER: UNSEEN: The likelihood is maximised for $N = n$. [1]

- (c) Comment on the merits of the Bayesian approach to a frequentist maximum likelihood estimator for this problem.

ANSWER: UNSEEN: The maximum likelihood estimator is the number of the taxi - i.e. it suggests that the taxi is the highest-numbered taxi in the city, which is implausible. The Bayesian posterior gives some probability for all $N \geq n$ up to the maximum number assumed. [3]

[9 marks]

[Total 25 marks]